MIL-M-28787/223A(EC)
12 March 1980
SUPERSEDING
MIL-M-28787/223(NAVY)
15 January 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE KDC

This specification is approved for use by the Naval Electronic System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital multiplexer module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table II.

Name	Miltiplexer
Family	Digital
Key code	KDC
Size	1A
Weight	36 grams maximum
Environment	Class I or class II, as specified in MIL-STD-1389

	Class I	Class II	Unit
Failure rate	0.181	0.522	Failures/10^6 hours maximum
Power dissipation	825 max	825 max	mW
Supply voltage	5.0 ±10	5.0 ±10	Vdc
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington D. C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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	Class I	Class II	Unit
Input current			
Low level (V_{IL} = 0.0 V)	-0.1 min, -1.73 max	-0.1 min, -1.73 max	mA
(V_{IL} = 0.4 V)	-0.1 min, -1.6 max	-0.1 min, -1.6 max	mA
High level (V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
(V_{IH} = 5.5 V)	0 min, 1.0 max	0 min, 1.0 max	mA
Output voltage			
Low level (I_{OL} = 16 mA)	0.0 min, 0.4 max	0.0 min, 0.4 max	V
High level (I_{OH} = -800 µA)	2.4 min, 4.5 max	2.4 min, 4.5 max	V
Output drive current			
Low level (V_{OL} = 0.4 V)	16 max	16 max	mA
High level (V_{OH} = 2.4 V)	-800 max	-800 max	µA
Propagation delay time			
t_{PLH}			
Select input to Y output			
C_L = 50 pF	6 min, 49 max	6 min, 58 max	ns
C_L = 150 pF	6 min, 59 max	6 min, 68 max	ns
Enable input to Y output			
C_L = 50 pF	3 min, 33 max	3 min, 37 max	ns
C_L = 150 pF	3 min, 43 max	3 min, 47 max	ns
Data input to Y output			
C_L = 50 pF	3 min, 35 max	3 min, 39 max	ns
C_L = 150 pF	3 min, 45 max	3 min, 49 max	ns
t_{PHL}			
Select input to Y output			
C_L = 50 pF	6 min, 41 max	6 min, 43 max	ns
C_L = 150 pF	6 min, 51 max	6 min, 53 max	ns
Enable input to Y output			
C_L = 50 pF	3 min, 39 max	3 min, 44 max	ns
C_L = 150 pF	3 min, 49 max	3 min, 54 max	ns
Data input to Y output			
C_L = 50 pF	3 min, 25 max	3 min, 27 max	ns
C_L = 150 pF	3 min, 35 max	3 min, 37 max	ns

1.3 Module function. This module contains two quad and one dual two input multiplexers. Each multiplexer receives two input data lines (D0, D1), an enable input ($\overline{E}$), and a select input (S). The data is transferred to the output (Y) when the input enable ($\overline{E}$) is low. Input D0 is selected when the select input (S) is low.

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1.3.1 Input to output relationship. Each multiplexer has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship for each multiplexer.

Enable	Select input	Inputs		Output
$\overline{E}$	S	D0	D1	Y
H	X	X	X	L
L	H	X	L	L
L	H	X	H	H
L	L	L	X	L
L	L	H	X	H

1/ H = high-voltage level.
2/ L = low-voltage level.
3/ X = don't care.

1.3.2 Parallel outputs. Paralleling of outputs is not permissible.

1.3.3 Crosstalk relationship. The module has the following characteristics.

1.3.3.1 Module crosstalk. When connected in the worst case mode of operation (all but one gate switching simultaneously from a high level to a low level) the KDC module could exhibit substantial crosstalk. When the quiet output is a low level, there may be as much as a 1,000 mV positive going spike coincident with the negative edge of the switching output.

1.3.3.2 Circuit crosstalk. Circuits 5 through 10 can induce as much as 400 mV of noise onto any circuit 1 through 4 output. Circuits 1 through 4, 9, and 10 can induce as much as 400 mV of noise onto any circuit 5 through 8 output. Circuits 1 through 8 can induce as much as 400 mV of noise onto any circuit 9 and 10 output.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specifiction for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

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3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.181 failures per million operating hours for class I or 0.522 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Pin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/223-1 for class I and M28787/223-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

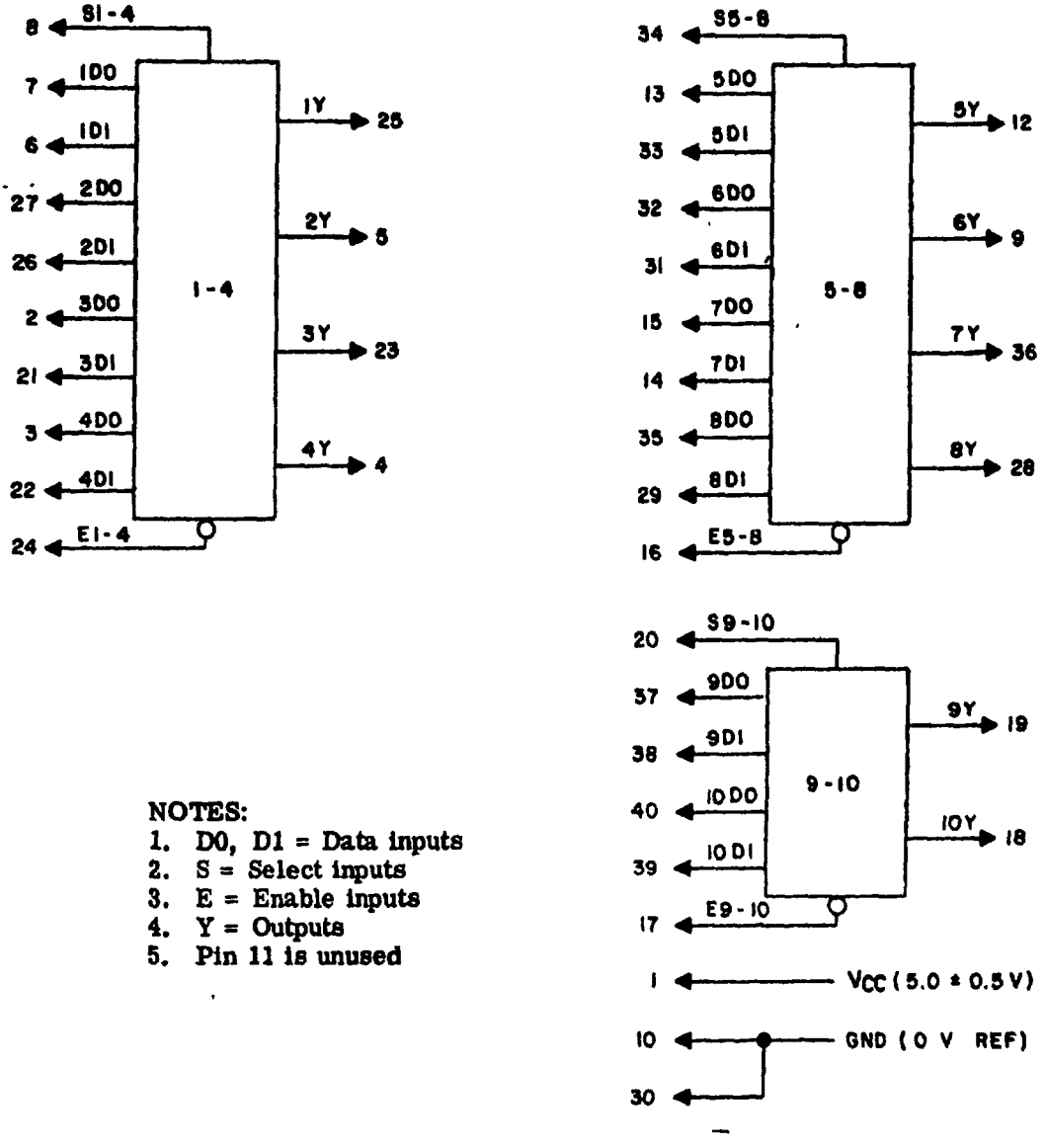
3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit apply only for a power supply voltage of 5.5 Vdc ±1 percent.

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NOTES:
1. D0, D1 = Data inputs
2. S = Select inputs
3. E = Enable inputs
4. Y = Outputs
5. Pin 11 is unused

FIGURE 1. KDC module function diagram, two input multiplexer.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/ 0° ±5°C		Class I 4/ 60° ±5°C		Class II 4/ -55° ±5°C		Class II 4/ 100° ±5°C		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation													
	Initial	III	S	10		10		10		10		10		MΩ
	End-of-life	III	S	1		1		1		1		1		MΩ
B1-B10	V_{OUT} (low-loaded)	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1-C26	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D10	Output short circuit current	VI	X	-20	-120	-20	-120	-20	-120	-20	-120	-20	-120	mA
E1-E26	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F5	Functional	VIII	X	← See table VIII →										
G1-G10	Static output voltage, low level	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H10	Static output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J26	Input current, high level	XI	X	0	40	0	40	0	40	0	40	0	40	μA
K1-K26	Input current, low level	XII	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA
L	Power supply current	XIII	S	10	150	10	150	10	150	10	150	10	150	mA
M1-M10	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	4	30	4	30	4	35	4	30	4	40	ns
M11-M20	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	4	40	4	40	4	45	4	40	4	50	ns
N1-N10	Transition time (t_{THL}), C_L = 50 pF	XIV	S	4	20	4	20	4	25	4	20	4	30	ns
N11-N20	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	4	25	4	25	4	30	4	25	4	35	ns

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/ 0° ±5°C		60° ±5°C		Class II 4/ -55° ±5°C		100° ±5°C		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P20	Propagation delay time (t_{PLH}) data to output, C_L = 50 pF	XIV	S	3	31	3	31	3	35	3	31	3	39	ns
P21-P40	Propagation delay time (t_{PLH}) data to output, C_L = 150 pF	XIV	Q	3	41	3	41	3	45	3	41	3	49	ns
Q1-Q10	Propagation delay time (t_{PLH}) select to output, C_L = 50 pF	XIV	S	6	40	6	40	6	49	6	40	6	58	ns
Q11-Q20	Propagation delay time (t_{PLH}) select to output, C_L = 150 pF	XIV	Q	6	50	6	50	6	59	6	50	6	68	ns
R1-R10	Propagation delay time (t_{PLH}) enable to output, C_L = 50 pF	XIV	S	3	29	3	29	3	33	3	29	3	37	ns
R11-R20	Propagation delay time (t_{PLH}) enable to output, C_L = 150 pF	XIV	Q	3	39	3	39	3	43	3	39	3	47	ns
S1-S20	Propagation delay time (t_{PHL}) data to output, C_L = 50 pF	XIV	S	3	23	3	23	3	25	3	23	3	27	ns
S21-S40	Propagation delay time (t_{PHL}) data to output, C_L = 150 pF	XIV	Q	3	33	3	33	3	35	3	33	3	37	ns
T1-T10	Propagation delay time (t_{PHL}) select to output, C_L = 50 pF	XIV	S	6	39	6	39	6	41	6	39	6	43	ns
T11-T20	Propagation delay time (t_{PHL}) select to output, C_L = 150 pF	XIV	Q	6	49	6	49	6	51	6	49	6	53	ns

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/ 0° ±5°C		60° ±5°C		Class II 4/ -55° ±5°C		100° ±5°C		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J10	Propagation delay time (tp_{HL}) enable to output, C_L = 50 pF	XIV	S	3	34	3	34	3	39	3	34	3	44	ns
J11-J20	Propagation delay time (tp_{HL}) enable to output, C_L = 150 pF	XIV	Q	3	44	3	44	3	49	3	44	3	54	ns
V1-V10	Output crosstalk, low level	XV	Q	0	1	0	1	0	1	0	1	0	1	V
V11-V20	Output crosstalk, low level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W10	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	uF
Z1-Z26	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	2	15	pF
AA1	Ground continuity	XIX	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
AA2	Ground continuity	XIX	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P07	4
Pulse generation	Item J01	1
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Capacitor meter	Item C05	1
Capacitance bridge	Item C01	1
Resistance bridge	Item R02	1
Resistors		
324Ω ±5%, 1/8 W		11
1.1Ω ±5%, 1 W		2
324Ω ±1%, 1/4 W		1
1.13kΩ±1%, 1/8 W		1
Value as required for figure 12, note 2, 1/8 W		10
3.01 kΩ ±1%, 1/8 W		10
Capacitors		
150 pF ±20%		10
Value as required for figure 12, note 4		10
Value as required for figure 12, note 5		10
Diode		
1N3600, 1N4150, or equivalent		10

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

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4.5 Qualification testing. Initial qualification tests and periodic-check test shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A thru AA in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through Y, and AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through Y and AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests E through AA shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/223-1 supersedes part number M28787/223. The two parts are interchangeable.

6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

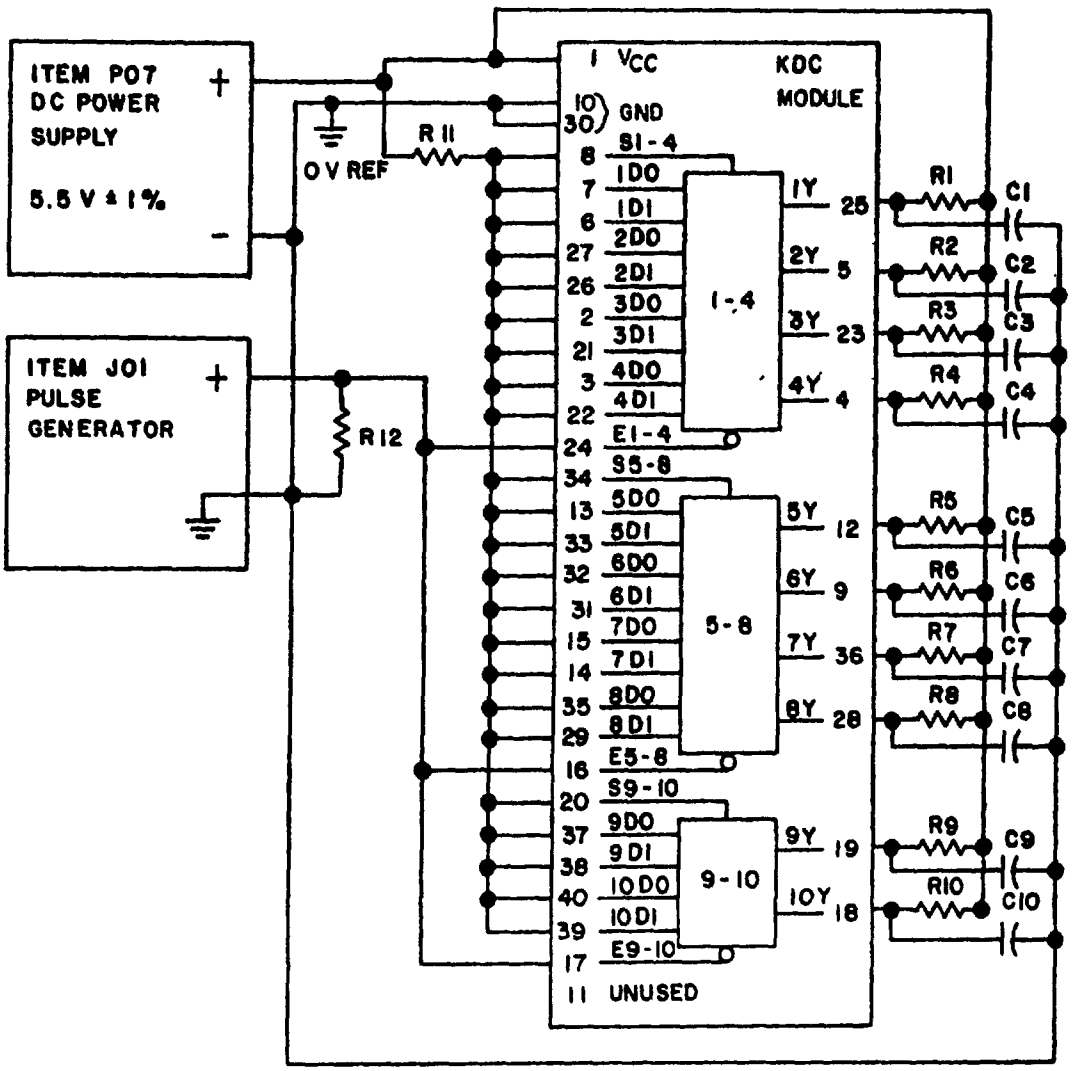
6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - EC

Agent:
DLA - ES

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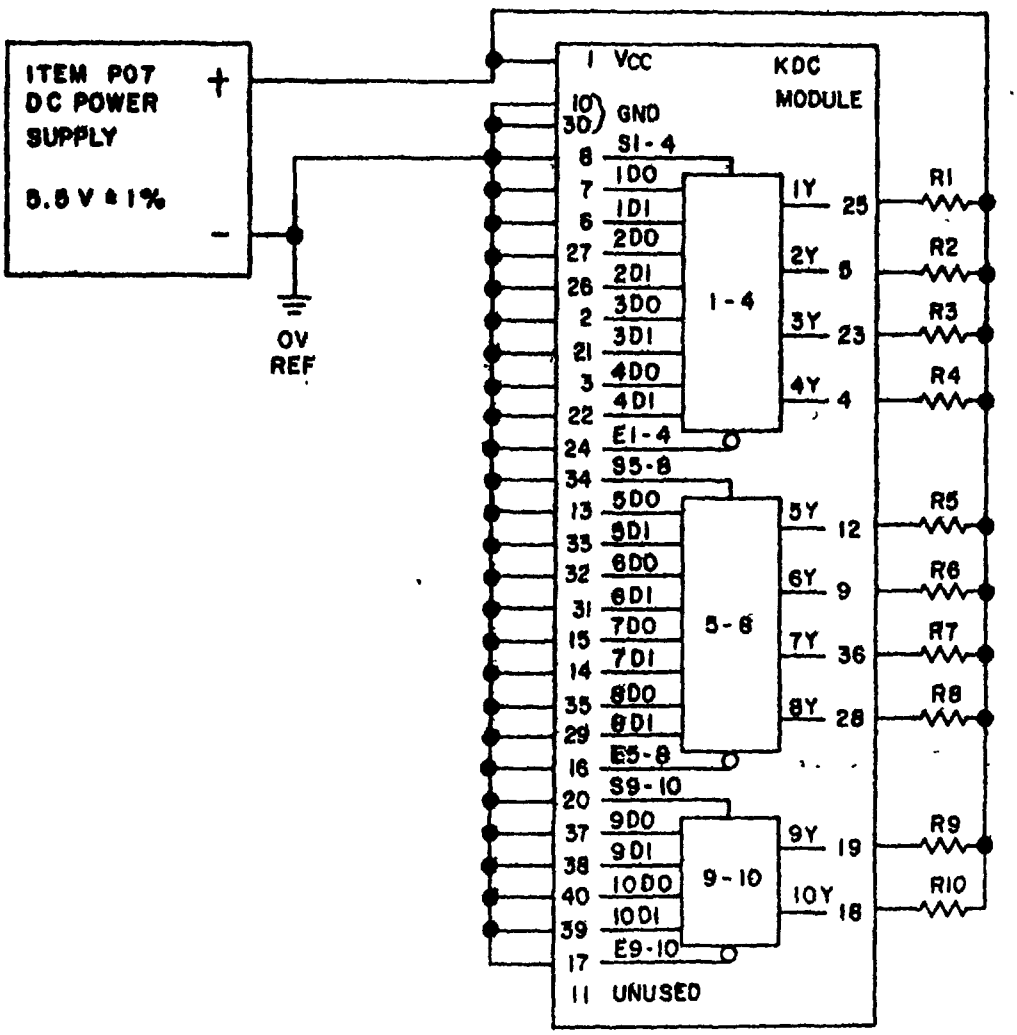
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NOTES:
1. R1-R11 = 324Ω ±5%, 1/8
2. R12 = 51.1Ω ±5%, 1 W.
3. C1-C10 = 150 pF ±20%.
4. Item J01 setup:
   PA = 4.0 ±1 V.
   PRR = 100 kHz, 50% duty cycle.

FIGURE 2. Operating temperature and life-test circuit.

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NOTE: R1-R10 = 324Ω ±5%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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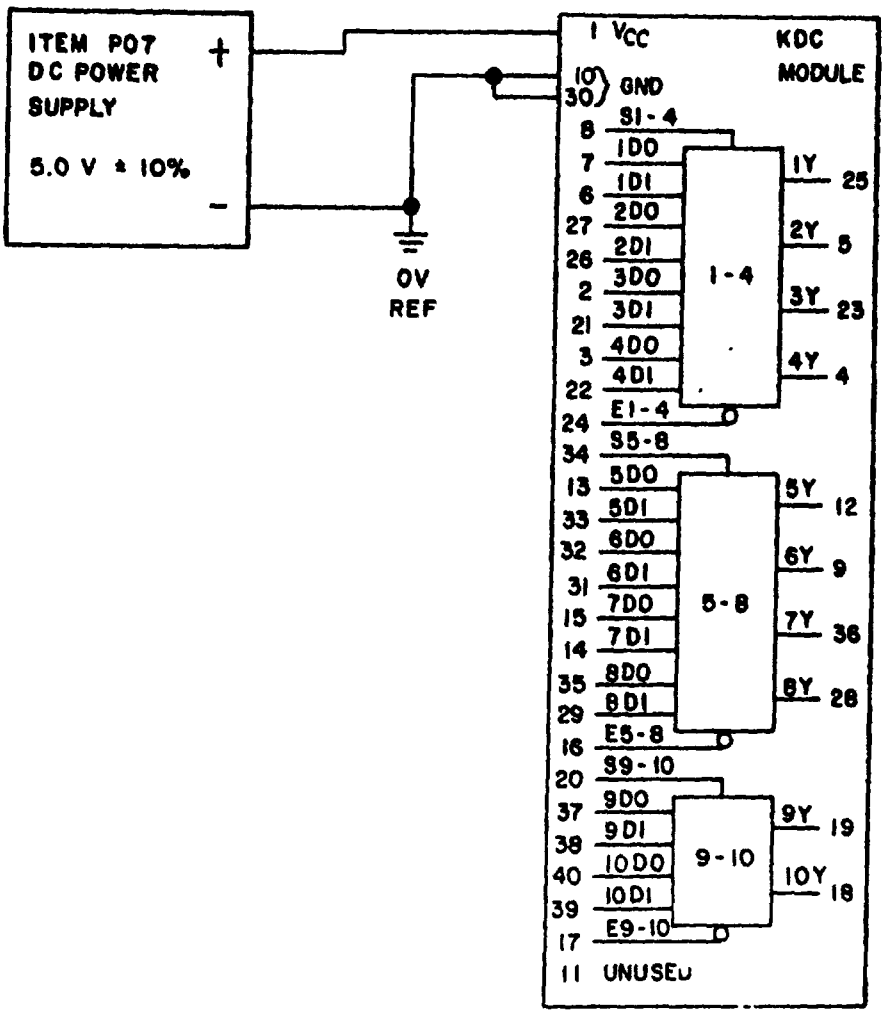


FIGURE 4. Vibration test circuit.

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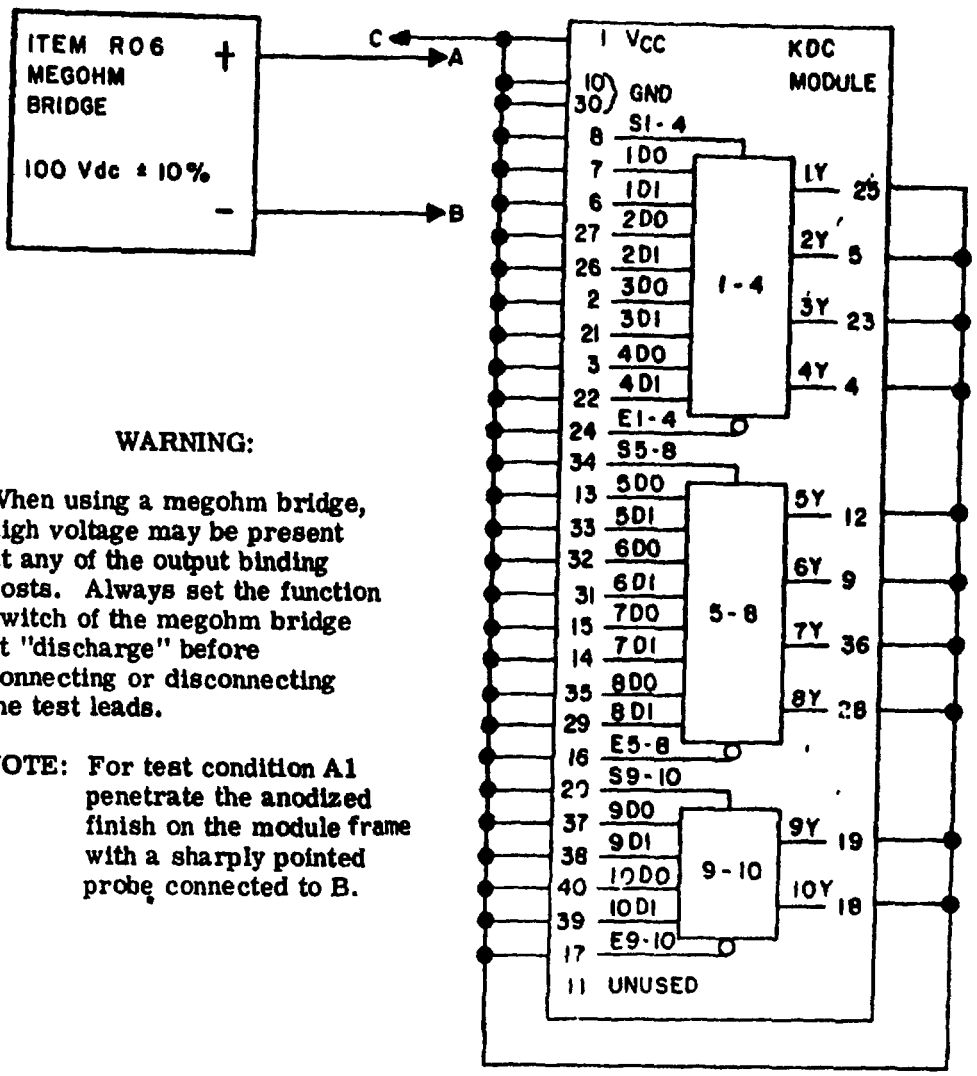
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 11	Frame	10		MΩ	5
A2	Isolation	C	11	10		MΩ	5

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1Y V_{OUT} (low-loaded)	25	0.01	0.8	V	6
B2	2Y	5				
B3	3Y	23				
B4	4Y	4				
B5	5Y	12				
B6	6Y	9				
B7	7Y	36				
B8	8Y	28				
B9	9Y	19				
B10	10Y	18				

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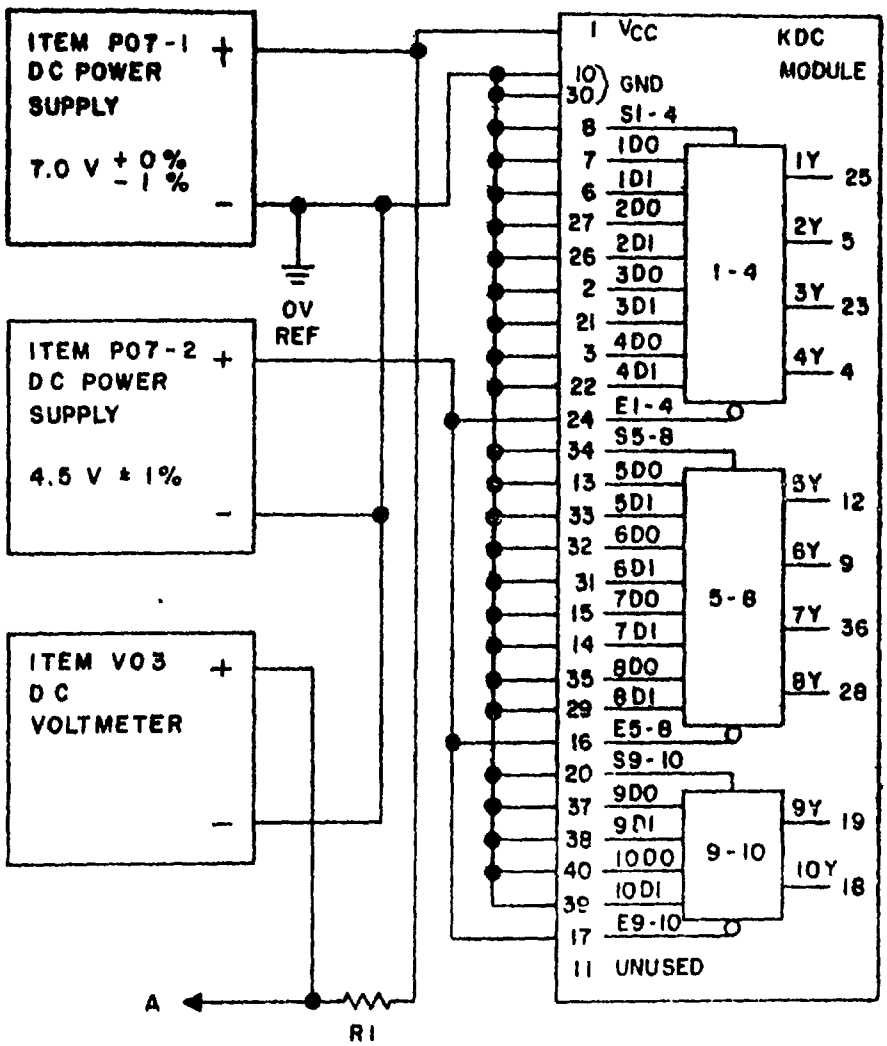
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to B.

FIGURE 5. Isolation test circuit for table III.

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NOTE: R1 = 324Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
C1	1D0 Input breakdown current	2,3,6,8,13-17,20-22,24,26,27, 29,31-35,37-40	7	0.0	1.0	mA	7
C2	1D1	2,3,7,8,13-17,20-22,24,26,27, 29,31-35,37-40	6				
C3	2D0	2,3,6-8,13-17,20-22,24,26,29, 31-35,37-40	27				
C4	2D1	2,3,6-8,13-17,20-22,24,27,29, 31-35,37-40	26				
C5	3D0	3,6-8,13-17,20-22,24,26,27, 29,31-35,37-40	2				
C6	3D1	2,3,6-8,13-17,20,22,24,26,27, 29,31-35,37-40	21				
C7	4D0	2,6-8,13-17,20-22,24,26,27, 29,31-35,37-40	3				
C8	4D1	2,3,6-8,13-17,20,21,24,26,27, 29,31-35,37-40	22				
C9	5D0	2,3,6-8,14-17,20-22,24,26,27, 29,31-35,37-40	13				
C10	5D1	2,3,6-8,13-17,20-22,24,26,27, 29,31,32,34,35,37-40	33				
C11	6D0	2,3,6-8,13-17,20-22,24,26,27, 29,31,33-35,37-40	32				
C12	6D1	2,3,6-8,13-17,20-22,24,26,27, 29,32-35,37-40	31				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ±5°C Ambient: Min	Initial limits 25° ±5°C Ambient: Max	Unit	Test figure
C13	7D0 Input breakdown current	2,3,6-8,13,14,16,17,20-22,24, 26,27,29,31-35,37-40	15	0.0	1.0	mA	7
C14	7D1	2,3,6-8,13,15-17,20-22,24,26, 27,29,31-35,37-40	14				
C15	8D0	2,3,6-8,13-17,20-22,24,26,27, 29,31-34,37-40	35				
C16	8D1	2,3,6-8,13-17,20-22,24,26,27, 31-35,37-40	29				
C17	9D0	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,38-40	37				
C18	9D1	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,37,39,40	38				
C19	10D0	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,37-39	40				
C20	10D1	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,37,38,40	39				
C21	S1-4	2,3,6,7,13-17,20-22,24,26,27, 29,31-35,37-40	8				
C22	E1-4	2,3,6-8,13-17,20-22,26,27,29, 31-35,37-40	24				
C23	S5-8	2,3,6-8,13-17,20-22,24,26,27, 29,31-33,35,37-40	34				

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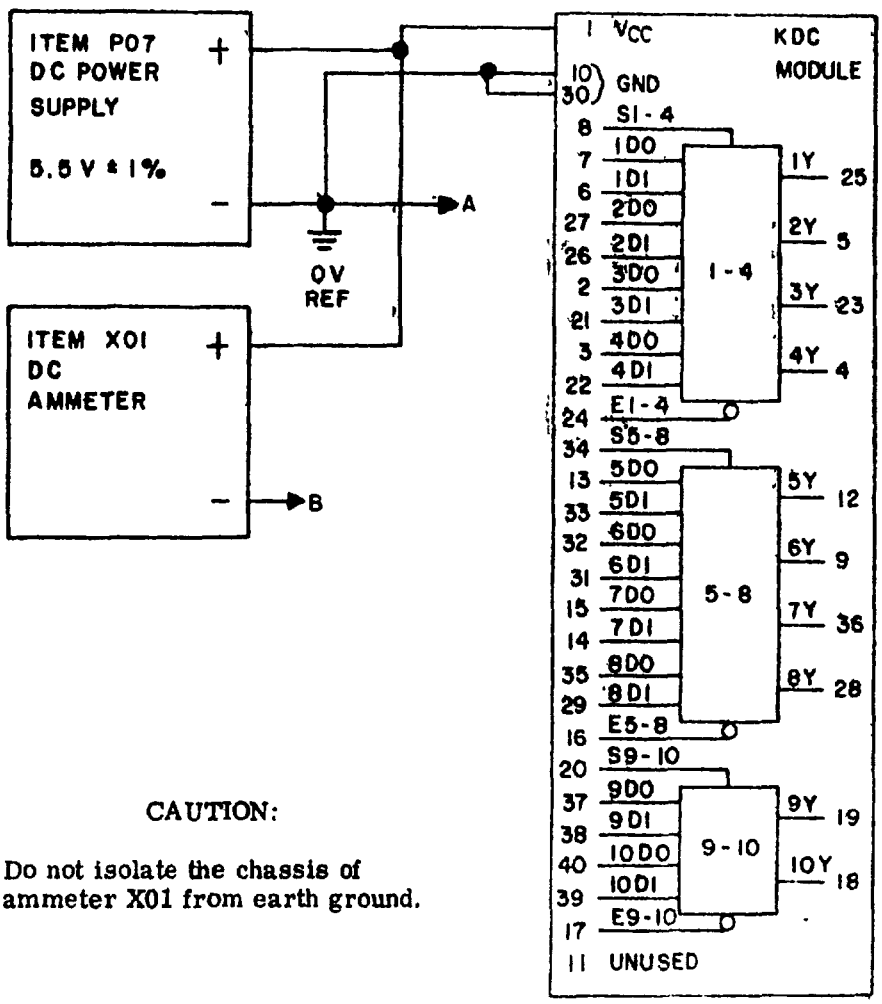
TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C24	E5-8 Input breakdown current	2,3,6-8,13-15,17,20-22,24,26, 27,29,31-35,37-40	16	0.0	1.0	mA	7
C25	S9-10 Input breakdown current	2,3,6-8,13-17,21,22,24,26,27, 29,31-35,37-40	20	0.0	1.0	mA	7
C26	E9-10 Input breakdown current	2,3,6-8,13-16,20-22,24,26,27, 29,31-35,37-40	17	0.0	1.0	mA	7

TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1Y Output short circuit current	25	-20	-120	mA	8
D2	2Y	5				
D3	3Y	23				
D4	4Y	4				
D5	5Y	12				
D6	6Y	9				
D7	7Y	36				
D8	8Y	28				
D9	9Y	19				
D10	10Y	18				

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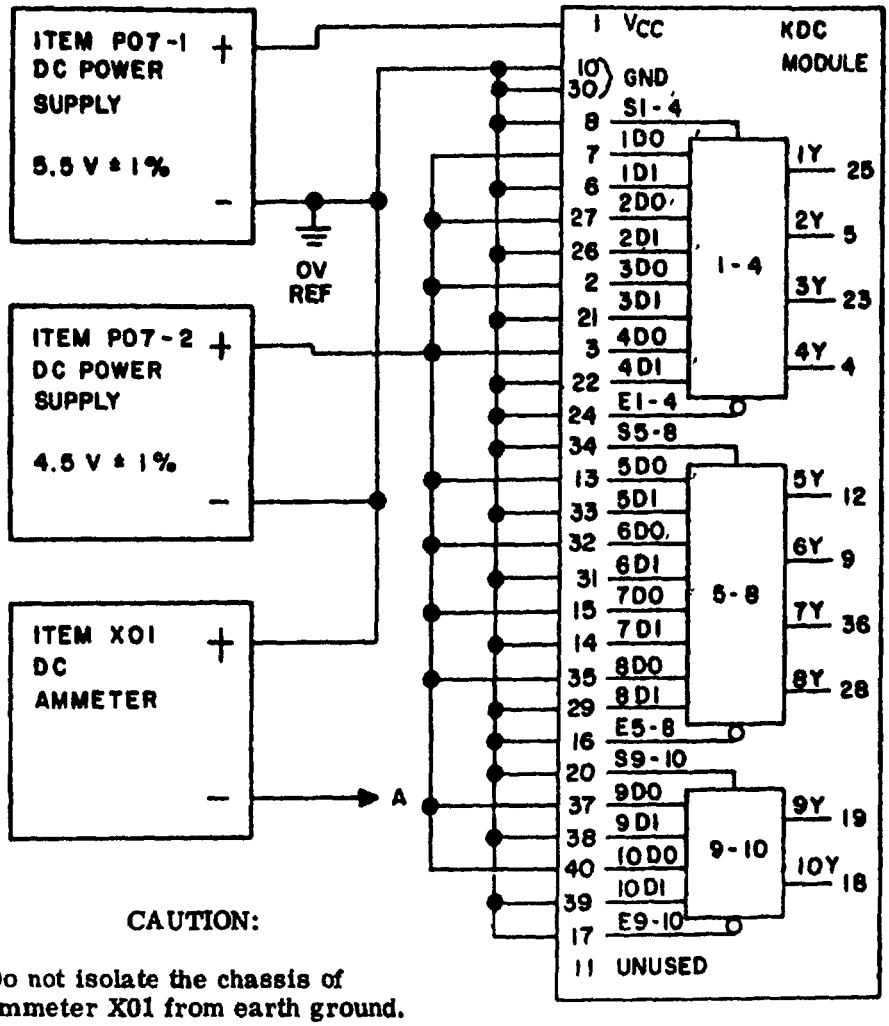


CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 7. Input breakdown current test for table V.

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NOTE: Only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
E1	S1-4 Input clamp diode voltage	8	-0.4	-1.5	V	9
E2	1D0	7				
E3	1D1	6				
E4	2D0	27				
E5	2D1	26				
E6	3D0	2				
E7	3D1	21				
E8	4D0	3				
E9	4D1	22				
E10	E1-4	24				
E11	S5-8	34				
E12	5D0	13				
E13	5D1	33				
E14	6D0	32				
F15	6D1	31				
E16	7D0	15				
E17	7D1	14				
E18	8D0	35				

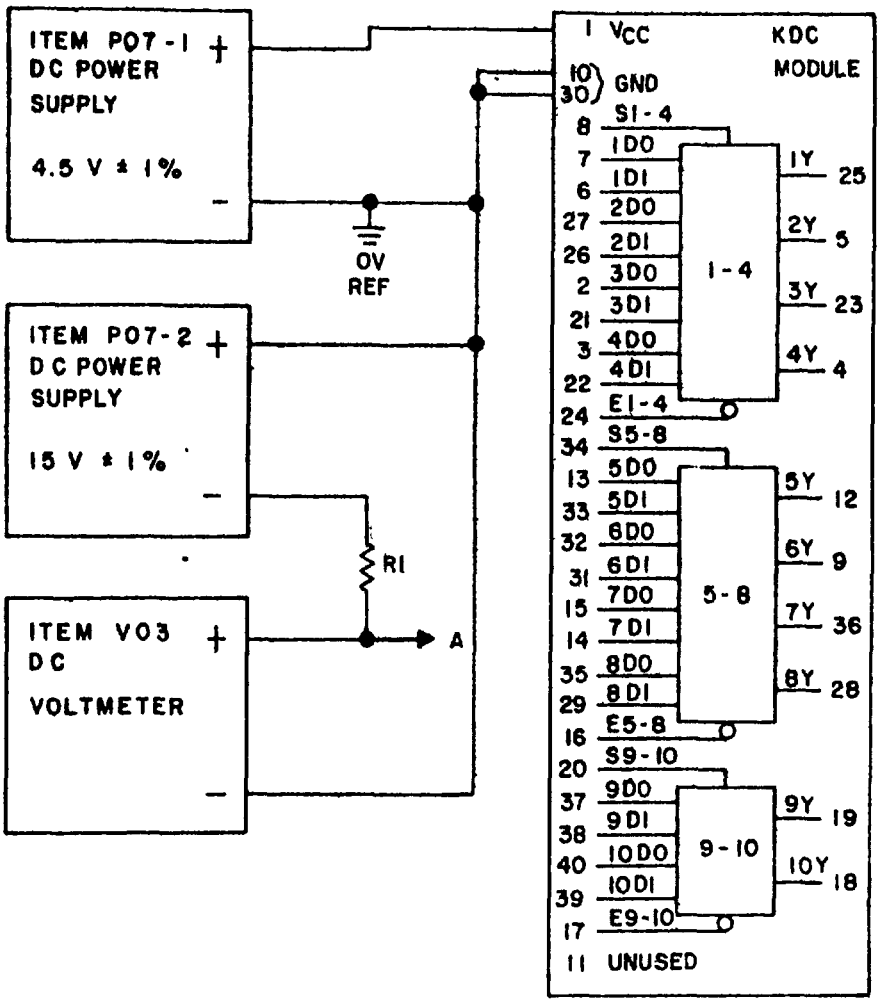
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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
E19	8D1 Input clamp diode voltage	29	-0.4	-1.5	V	9
E20	E5-8	16				
E21	S9-10	20				
E22	9D0	37				
E23	9D1	38				
E24	10D0	40				
E25	10D1	39				
E26	E9-10	17				

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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

FIGURE 9. Input clamp diode voltage test circuit for table VII.

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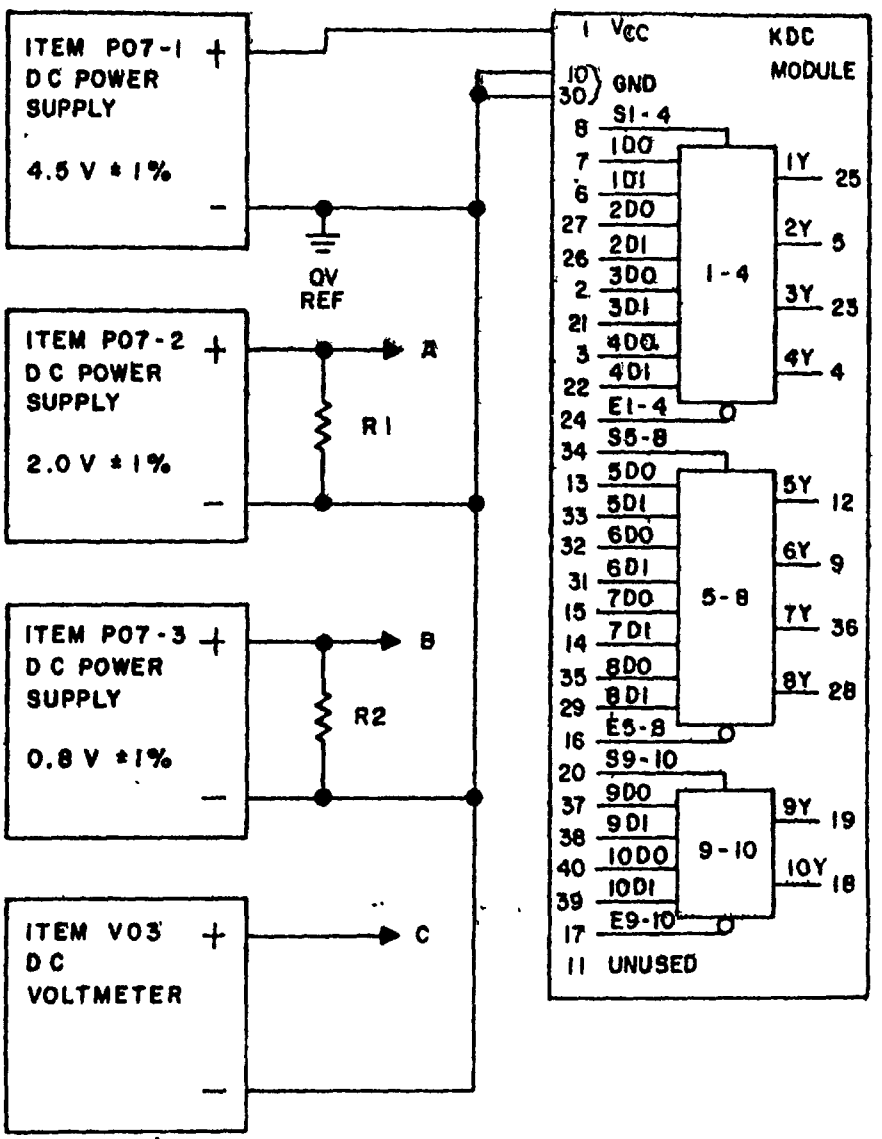
TABLE VIII. Functional test.

Test condition	Parameter	Connect			Output logic states										Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	7Y	8Y	9Y	10Y	Min	Max		
F1	Functional with E=H S=H, D0=H, D1=H	16,17,24,8, 34,20,7,27, 2,3,13,32, 15,35,37,40, 6,26,21,22, 33,31,14,29, 38,39	Open	See notes 2 & 3	L	L	L	L	L	L	L	L	L	L	See note 1			10
F2	Functional with E=L S=H, D0=H, D1=L	8,34,20,7, 27,2,3,13, 32,15,35, 37,40	6,26,21,22, 33,31,14, 16,17,24, 29,38,39		L	L	L	L	L	L	L	L	L	L				
F3	Functional with E=L, S=H, D0=L, D1=H	8,34,20,6, 26,21,22, 33,31,14, 29,38,39	7,27,2,3,13, 32,15,35, 16,17,24, 37,40		H	H	H	H	H	H	H	H	H	H				
F4	Functional with E=L, S=L, D0=L, D1=L	6,20,21,22, 33,31,14, 29,38,39	16,17,24,8, 34,20,7,27, 2,3,13,32 15,35,37,40		L	L	L	L	L	L	L	L	L	L				
F5	Functional with E=L, S=L, D0=H, D1=L	7,27,2,3, 13,32,15, 35,37,40	16,17,24,8, 34,20,6,26, 21,22,33,31, 14,29,38,39		H	H	H	H	H	H	H	H	H	H				

NOTES:

1. Output logic states: L = Low-voltage level = 0.0 to 0.4 V.
   H = High-voltage level = 2.4 to 4.5 V.
2. For test conditions F1-F5, verification of the output logic states in note 1 is obtained by connecting item V03 (point C) to each output pin in succession after input conditions have been accomplished.
3. 1Y through 10Y output states are measured at output pins 25, 5, 23, 4, 12, 9, 36, 28, 19, and 18 respectively.

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NOTE: R1 = R2 = 51.1Ω ±5%, 1 W.

FIGURE 10. Functional test circuit for table VIII.

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TABLE IX. Static output voltage (low-level) test.

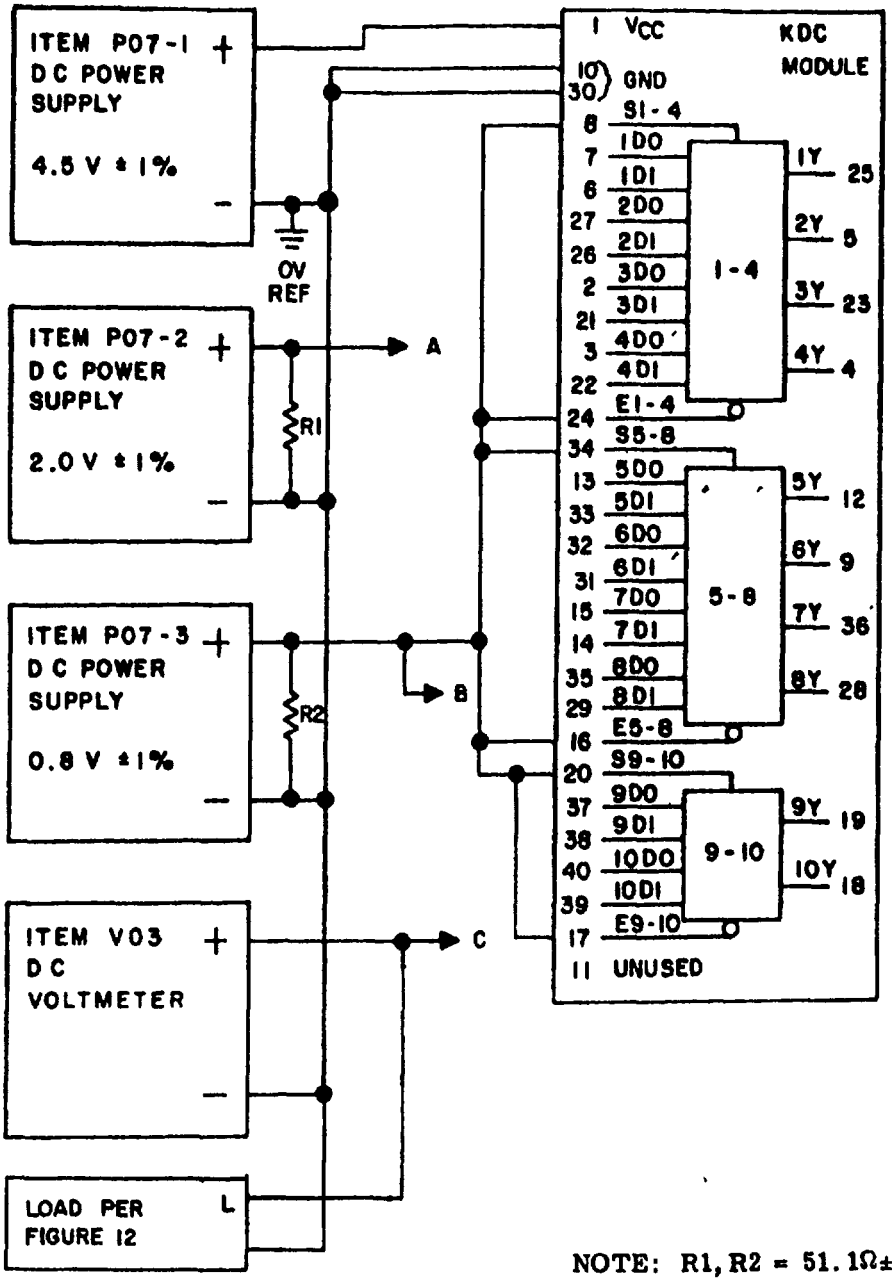
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	1Y Static output voltage, low level	Open	7	25	0.0	0.4	V	11
C2	2Y		27	5				
C3	3Y		2	23				
C4	4Y		3	4				
C5	5Y		13	12				
C6	6Y		32	9				
C7	7Y		15	36				
C8	8Y		35	28				
C9	9Y		37	19				
C10	10Y		40	18				

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TABLE X. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1Y Static output voltage, high level	7	Open	25	2.4	4.5	V	11
H2	2Y	27		5				
H3	3Y	2		23				
H4	4Y	3		4				
H5	5Y	13		12				
H6	6Y	32		9				
H7	7Y	15		36				
H8	8Y	35		28				
H9	9Y	37		19				
H10	10Y	40		18				

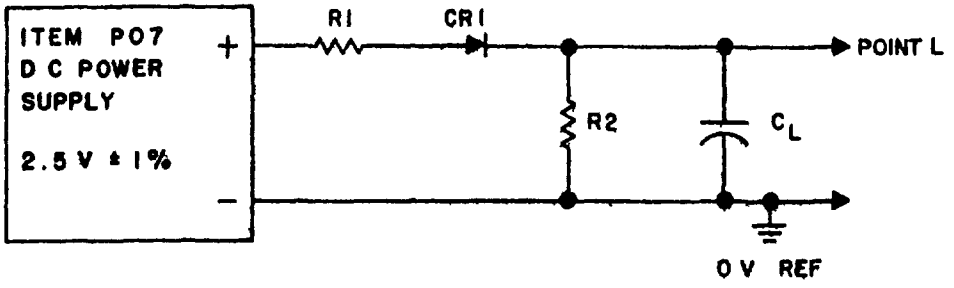
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NOTE: R1, R2 = 51.1Ω ± 5℃, 1W.

FIGURE 11. Static output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 3.01 kΩ ±1%, 1/8 W.
4. For test conditions M1-M10, N1-N10, P1-P20, Q1-Q10, R1-R10, S1-S20, T1-T10, U1-U10, V1-V20, and W1-W10: C_L = 50 ±1 pF (including probe and fixture capacitance).
5. For test conditions M11-M20, N11-N20, P21-P40, Q11-Q20, R11-R20, S21-S40, T11-T20, and U11-U20: C_L = 150 ±5 pF (including probe and fixture capacitance).
6. Capacitor C_L is not required for dc electrical test conditions G1-G10 and H1-H10.
7. Lead length from load circuit to module output pin under test is to be less than 3 inches (76.2 mm).

FIGURE 12. KDC logic load circuit.

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TABLE XI. Input current (high-level) test.

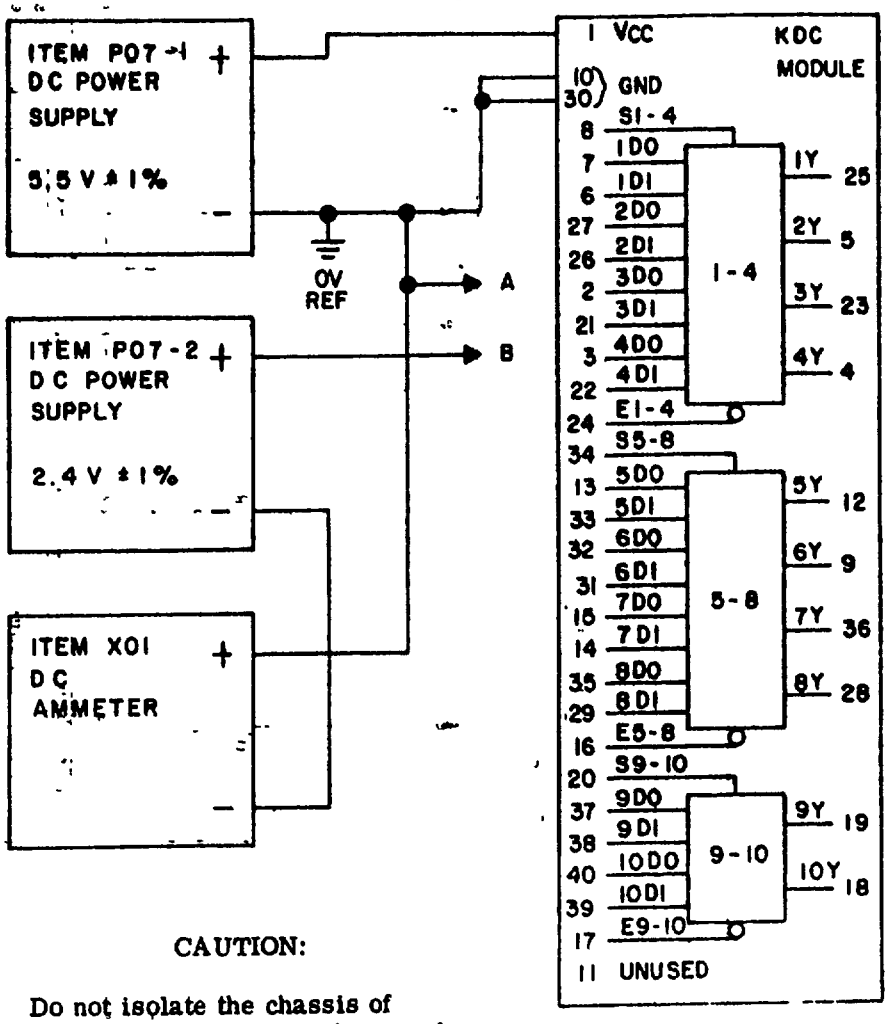
Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
I1	1D0 Input current, high level	2,3,6,8,13-17,20-22,24,26,27, 29,31-35,37-40	7	0	40	µA	13
I2	1D1	2,3,7,8,13-17,20-22,24,26,27, 29,31-35,37-40	6				
I3	2D0	2,3,6-8,13-17,20-22,24,26,29, 31-35,37-40	27				
I4	2D1	2,3,6-8,13-17,20-22,24,27,29, 31-35,37-40	26				
I5	3D0	3,6-8,13-17,20-22,24,26,27,29, 31-35,37-40	2				
I6	3D1	2,3,6-8,13-17,20,22,24,26,27, 29,31-35,37-40	21				
I7	4D0	2,6-8,13-17,20-22,24,26,27,29, 31-35,37-40	3				
I8	4D1	2,3,6-8,13-17,20,21,24,26,27, 29,31-35,37-40	22				
I9	5D0	2,3,6-8,14-17,20-22,24,26,27, 29,31-35,37-40	13				
I10	5D1	2,3,6-8,13-17,20-22,24,26,27, 29,31,32,34,35,37-40	33				
I11	6D0	2,3,6-8,13-17,20-22,24,26,27, 29,31,33-35,37-40	32				
I12	6D1	2,3,6-8,13-17,20-22,24,26,27, 29,32-35,37-40	31				
I13	7D0	2,3,6-8,13,14,16,17,20-22,24, 26,27,29,31-35,37-40	15				

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J14	/D1 input current, high level	2,3,6-8,13,15-17,20-22,24,26, 27,29,31-35,37-40	14	0	40	µA	13
J15	8D0	2,3,6-8,13-17,20-22,24,26,27, 29,31-34,37-40	35				
J16	8D1	2,3,6-8,13-17,20-22,24,26,27, 31-35,37-40	29				
J17	9D0	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,38-40	37				
J18	9D1	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,37,39,40	38				
J19	10D0	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,37-39	40				
J20	10D1	2,3,6-8,13-17,20-22,24,26,27, 29,31-35,37,38,40	39				
J21	S1-4	2,3,6,7,13-17,20-22,24,26,27, 29,31-35,37-40	8				
J22	E1-4	2,3,6-8,13-17,20-22,26,27,29, 31-35,37-40	24				
J23	S5-8	2,3,6-8,13-17,20-22,24,26,27, 29,31-33,35,37-40	34				
J24	E5-8	2,3,6-8,13-15,17,20-22,24,26, 27,29,31-35,37-40	16				
J25	S9-10	2,3,6-8,13-17,21,22,24,26,27, 29,31-35,37-40	20				
J26	E9-10	2,3,6-8,13-16,20-22,24,26,27, 29,31-35,37-40	17				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 13. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	1D0	Input current, low level	8,24	7	-0.1	-1.73	mA	14
K2	1D1		24	6				
K3	2D0		8,24	27				
K4	2D1		24	26				
K5	3D0		8,24	2				
K6	3D1		24	21				
K7	4D0		8,24	3				
K8	4D1		24	22				
K9	5D0		34,16	13				
K10	5D1		16	33				
K11	6D0		34,16	32				
K12	6D1		16	31				
K13	7D0		34,16	15				
K14	7D1		16	14				
K15	8D0		34,16	35				
K16	8D1		16	29				
K17	9D0		20,17	37				

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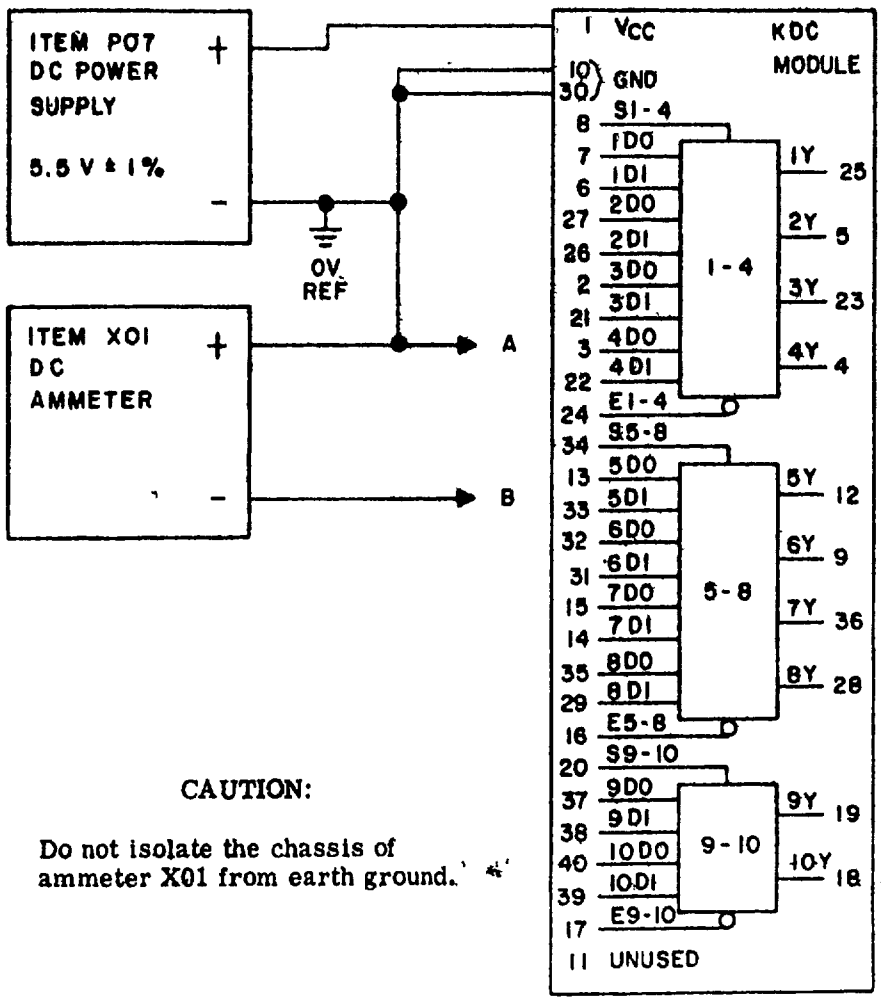
TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K18	9D1 Input current, low level	17	38	-0.1	-1.73	mA	14
K19	10D0	20,17	40				
K20	10D1	17	39				
K21	S1-4	Open	8				
K22	E1 4		24				
K23	S5-8		34				
K24	E5-8		16				
K25	S9-10		20				
K26	E9-10		17				

TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
1.	Power supply current, low-level inputs	10	150	mA	15

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 14 Input current (low-level) test circuit for table XII.

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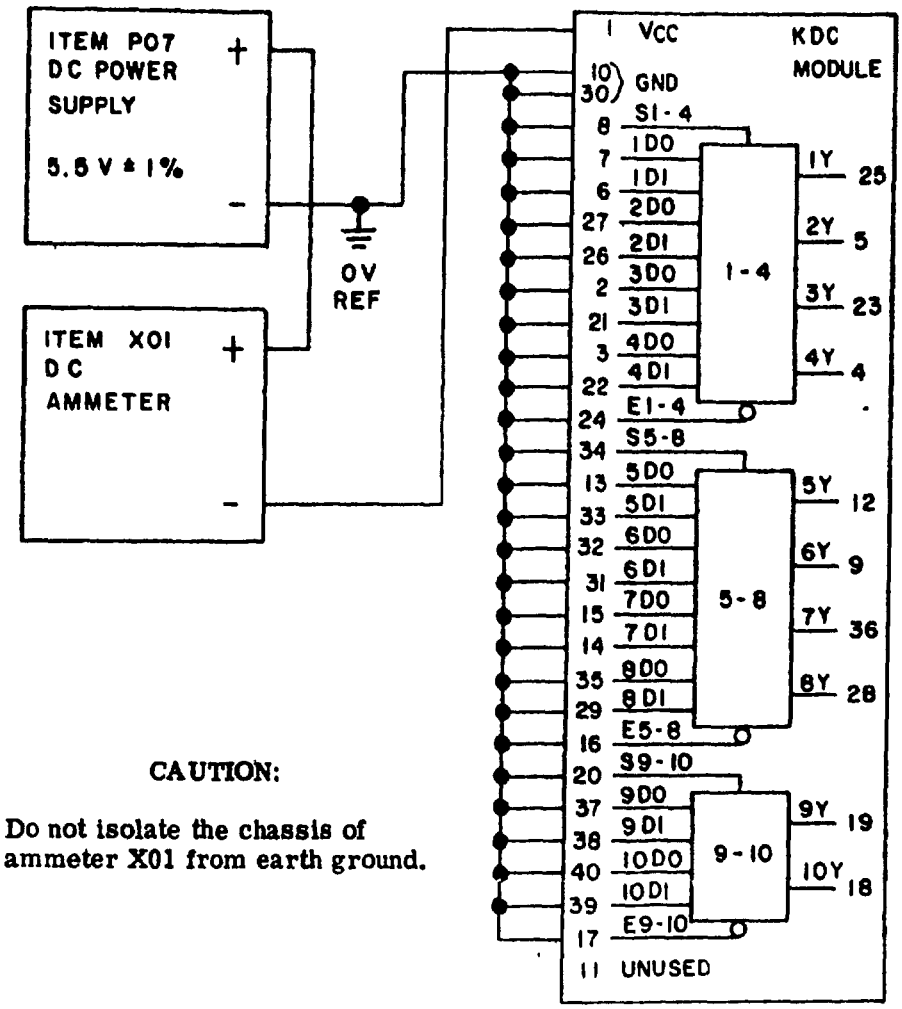


FIGURE 15. Power supply current test circuit for table XIII.

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TABLE XIV. Transition time and propagation delay time test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1Y Transition time, (t_{TLH}), C_L = 50 pF	Open	8,24	7	24	4	30	ns	16,17
M2	2Y		8,24	27	5				
M3	3Y		8,24	2	23				
M4	4Y		8,24	3	4				
M5	5Y		34,16	13	12				
M6	6Y		34,16	32	9				
M7	7Y		34,16	15	36				
M8	8Y		34,16	35	28				
M9	9Y		17,20	37	19				
M10	10Y		17,20	40	18				
M11	1Y Transition time (t_{TLH}), C_L = 150 pF		8,24	7	25		40		
M12	2Y		8,24	27	5				
M13	3Y		8,24	2	23				
M14	4Y		8,24	3	4				
M15	5Y		34,16	13	12				
M16	6Y		34,16	32	9				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M17	7Y Transition time (t_{TLH}), C_L = 150 pF	Open	34,16	15	36	4	40	ns	16,17
M18	8Y		34,16	35	28				
M19	9Y		17,20	37	19				
M20	10Y		17,20	40	40				
N1	1Y Transition time (t_{THL}), C_L = 50 pF		8,24	7	25		20		
N2	2Y		8,24	27	5				
N3	3Y		8,24	2	23				
N4	4Y		8,24	3	4				
N5	5Y		34,16	13	12				
N6	6Y		34,16	32	9				
N7	7Y		34,16	15	36				
N8	8Y		34,16	35	28				
N9	9Y		17,20	37	19				
N10	10Y		17,20	40	18				
N11	1Y Transition time (t_{THL}), C_L = 150 pF		8,24	7	25		25		

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N12	2Y Transition time (t_{THL}), C_L = 150 pF	Open	8,24	27	5	4	25	ns	16,17
N13	3Y		8,24	2	23				
N14	4Y		8,24	3	4				
N15	5Y		34,16	13	12				
N16	6Y		34,16	32	9				
N17	7Y		34,16	15	36				
N18	8Y		34,16	35	28				
N19	9Y		17,20	37	19				
N20	10Y		17,20	40	18				
P1	1DO to 1Y Propagation delay time (t_{PLH}), C_L = 50 pF		8,24	7	25	3	31		16,18
P2	2DO to 2Y		8,24	27	5				
P3	3DO to 3Y		8,24	2	23				
P4	4DO to 4Y		8,24	3	4				
P5	5DO to 5Y		16,34	13	12				
P6	6DO to 6Y		16,34	32	9				
P7	7DO to 7Y		16,34	15	36				

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TABLE XIV. Transition time and propagation delay time - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P8	8D0 to 8Y Propagation delay time (t_{PLH}), C_L = 50 pF	Open	16,34	35	28	3	31	ns	16,18
P9	9D0 to 9Y	Open	17,20	37	19				
P10	10D0 to 10Y	Open	17,20	40	18				
P11	1D1 to 1Y	8	24	6	25				
P12	2D1 to 2Y	8	24	26	5				
P13	3D1 to 3Y	8	24	21	23				
P14	4D1 to 4Y	8	24	22	4				
P15	5D1 to 5Y	34	16	33	12				
P16	6D1 to 6Y	34	16	31	9				
P17	7D1 to 7Y	34	16	14	36				
P18	8D1 to 8Y	34	16	29	28				
P19	9D1 to 9Y	20	17	38	19				
P20	10D1 to 10Y	20	17	39	18				
P21	1D0 to 1Y Propagation delay time (t_{PLH}), C_L = 150 pF	Open	8,24	7	25		41		
P22	2D0 to 2Y Propagation delay time (t_{PLH}), C_L = 150 pF	Open	8,24	27	5		41		

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P23	3D0 to 3Y Propagation delay time (t_{PLH}), C_L = 150 pF	Open	8,24	2	23	3	41	ns	16,18
P24	4D0 to 4Y		8,24	3	4				
P25	5D0 to 5Y		16,34	13	12				
P26	6D0 to 6Y		16,34	32	9				
P27	7D0 to 7Y		16,34	15	36				
P28	8D0 to 8Y		16,34	35	28				
P29	9D0 to 9Y		17,20	37	19				
P30	10D0 to 10Y		17,20	40	18				
P31	1D1 to 1Y	8	24	6	25				
P32	2D1 to 2Y	8	24	26	5				
P33	3D1 to 3Y	8	24	21	23				
P34	4D1 to 4Y	8	24	22	4				
P35	5D1 to 5Y	34	16	33	12				
P36	6D1 to 6Y	34	16	31	9				
P37	7D1 to 7Y	34	16	14	36				
P38	8D1 to 8Y	34	16	29	28				
P39	9D1 to 9Y	20	17	38	19				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P40	10D1 to 10Y Propagation delay time (t_{PLH}), C_L = 150 pF	20	17	39	18	3	41	ns	16,18
Q1	S1-4 to 1Y Propagation delay time (t_{PLH}), C_L = 50 pF	6	7,24	8	25	6	40		
Q2	S1-4 to 2Y	26	27,24	8	5				
Q3	S1-4 to 3Y	21	2,24	8	23				
Q4	S1-4 to 4Y	22	3,24	8	4				
Q5	S5-8 to 5Y	33	13,16	34	12				
Q6	S5-8 to 6Y	31	32,16	34	9				
Q7	S5-8 to 7Y	14	15,16	34	36				
Q8	S5-8 to 8Y	29	35,16	34	28				
Q9	S9-10 to 9Y	38	37,17	20	19				
Q10	S9-10 to 10Y	39	40,17	20	18				
Q11	S1-4 to 1Y Propagation delay time (t_{PLH}), C_L = 150 pF	6	7,24	8	25		50		
Q12	S1-4 to 2Y Propagation delay time (t_{PLH}), C_L = 150 pF	26	27,24	8	5				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q13	S1-4 to 3Y Propagation delay time (t_{PLH}), C_L = 150 pF	21	2,24	8	23	6	50	ns	16,18
Q14	S1-4 to 4Y	22	3,24	8	4				
Q15	S5-8 to 5Y	33	13,16	34	12				
Q16	S5-8 to 6Y	31	32,16	34	9				
Q17	S5-8 to 7Y	14	15,16	34	36				
Q18	S5-8 to 8Y	29	35,16	34	28				
Q19	S9-10 to 9Y	38	37,17	20	19				
Q20	S9-10 to 10Y	39	40,17	20	18				
R1	E1-4 to 1Y Propagation delay time (t_{PLH}), C_L = 50 pF	7	8	24	25	3	29		16,19
R2	E1-4 to 2Y	27	8	24	5				
R3	E1-4 to 3Y	2	8	24	23				
R4	E1-4 to 4Y	3	8	24	4				
R5	E5-8 to 5Y	13	34	16	12				
R6	E5-8 to 6Y	32	34	16	9				
R7	E5-8 to 7Y	15	34	16	36				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R8	E5-8 to 8Y Propagation delay time (t_{PLH}), C_L = 50 pF	35	34	16	28	3	29	ns	16,19
R9	E9-10 to 9Y Propagation delay time (t_{PLH}), C_L = 50 pF	37	20	17	19		29		
R10	E9-10 to 10Y Propagation delay time (t_{PLH}), C_L = 50 pF	40	20	17	18		29		
R11	E1-4 to 1Y Propagation delay time (t_{PLH}), C_L = 150 pF	7	8	24	25		39		
R12	E1-4 to 2Y	27	8	24	5				
R13	E1-4 to 3Y	2	8	24	23				
R14	E1-4 to 4Y	3	8	24	4				
R15	E5-8 to 5Y	13	34	16	12				
R16	E5-8 to 6Y	32	34	16	9				
R17	E5-8 to 7Y	15	34	16	36				
R18	E5-8 to 8Y	35	34	16	28				
R19	E9-10 to 9Y	37	20	17	19				
R20	E9-10 to 10Y	40	20	17	18				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1D0 to 1Y Propagation delay time (t_{PHL}), C_L = 50 pF	Open	8,24	7	25	3	23	ns	16,18
S2	2D0 to 2Y		8,24	27	5				
S3	3D0 to 3Y		8,24	2	23				
S4	4D0 to 4Y		8,24	3	4				
S5	5D0 to 5Y		16,34	13	12				
S6	6D0 to 6Y		16,34	32	9				
S7	7D0 to 7Y		16,34	15	36				
S8	8D0 to 8Y		16,34	35	28				
S9	9D0 to 9Y		17,20	37	19				
S10	10D0 to 10Y		17,20	40	18				
S11	1D1 to 1Y	8	24	6	25				
S12	2D1 to 2Y	8	24	26	5				
S13	3D1 to 3Y	8	24	21	23				
S14	4D1 to 4Y	8	24	22	4				
S15	5D1 to 5Y	34	16	33	12				
S16	6D1 to 6Y	34	16	31	9				
S17	7D1 to 7Y	34	16	14	36				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S18	8D1 to 8Y Propagation delay time (t_{PHL}), C_L = 50 pF	34	16	29	28	3	23	ns	16,18
S19	9D1 to 9Y Propagation delay time (t_{PHL}), C_L = 50 pF	20	17	38	19				
S20	10D1 to 10Y Propagation delay time (t_{PHL}), C_L = 50 pF	20	17	39	18				
S21	1D0 to 1Y Propagation delay time (t_{PHL}), C_L = 150 pF	Open	8,24	7	25		33		
S22	2D0 to 2Y		8,24	27	5				
S23	3D0 to 3Y		8,24	2	23				
S24	4D0 to 4Y		8,24	3	4				
S25	5D0 to 5Y		16,34	13	12				
S26	6D0 to 6Y		16,34	32	9				
S27	7D0 to 7Y		16,34	15	36				
S28	8D0 to 8Y		16,34	35	28				
S29	9D0 to 9Y		17,20	37	19				
S30	10D0 to 10Y		17,20	40	18				
S31	1D1 to 1Y	8	24	6	25				
S32	2D1 to 2Y	8	24	26	5				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S33	3D1 to 3Y Propagation delay time (t_{PHL}), C_L = 150 pF	8	24	21	23	3	33	ns	16,18
S34	4D1 to 4Y	8	24	22	4				
S35	5D1 to 5Y	34	16	33	12				
S36	6D1 to 6Y	34	16	31	9				
S37	7D1 to 7Y	34	16	14	36				
S38	8D1 to 8Y	34	16	29	28				
S39	9D1 to 9Y	20	17	38	19				
S40	10D1 to 10Y	20	17	39	18				
T1	S1-4 to 1Y Propagation delay time (t_{PHL}), C_L = 50 pF	6	7,24	8	25	6	39		
T2	S1-4 to 2Y	26	27,24	8	5				
T3	S1-4 to 3Y	21	2,24	8	23				
T4	S1-4 to 4Y	22	3,24	8	4				
T5	S5-8 to 5Y	33	13,16	34	12				
T6	S5-8 to 6Y	31	32,16	34	9				
T7	S5-8 to 7Y	14	15,16	34	36				
T8	S5-8 to 8Y	29	35,16	34	28				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T9	S9-10 to 9Y Propagation delay time (t_{PHL}), C_L = 50 pF	38	37,17	20	19	6	39	ns	16,18
T10	S9-10 to 10Y Propagation delay time (t_{PHL}), C_L = 50 pF	39	40,17	20	18		39		
T11	S1-4 to 1Y Propagation delay time (t_{PHL}), C_L = 150 pF	6	7,24	8	25		49		
T12	S1-4 to 2Y	26	27,24	8	5				
T13	S1-4 to 3Y	21	2,24	8	23				
T14	S1-4 to 4Y	22	3,24	8	4				
T15	S5-8 to 5Y	33	13,16	34	12				
T16	S5-8 to 6Y	31	32,16	34	9				
T17	S5-8 to 7Y	14	15,16	34	36				
T18	S5-8 to 8Y	29	35,16	34	28				
T19	S9-10 to 9Y	38	37,17	20	19				
T20	S9-10 to 10Y	39	40,17	20	18				
U1	E1-4 to 1Y Propagation delay time (t_{PHL}), C_L = 50 pF	7	8	24	25	3	34		16,19

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TABLE XIV. Transition time and propagation delay time test - Continued.

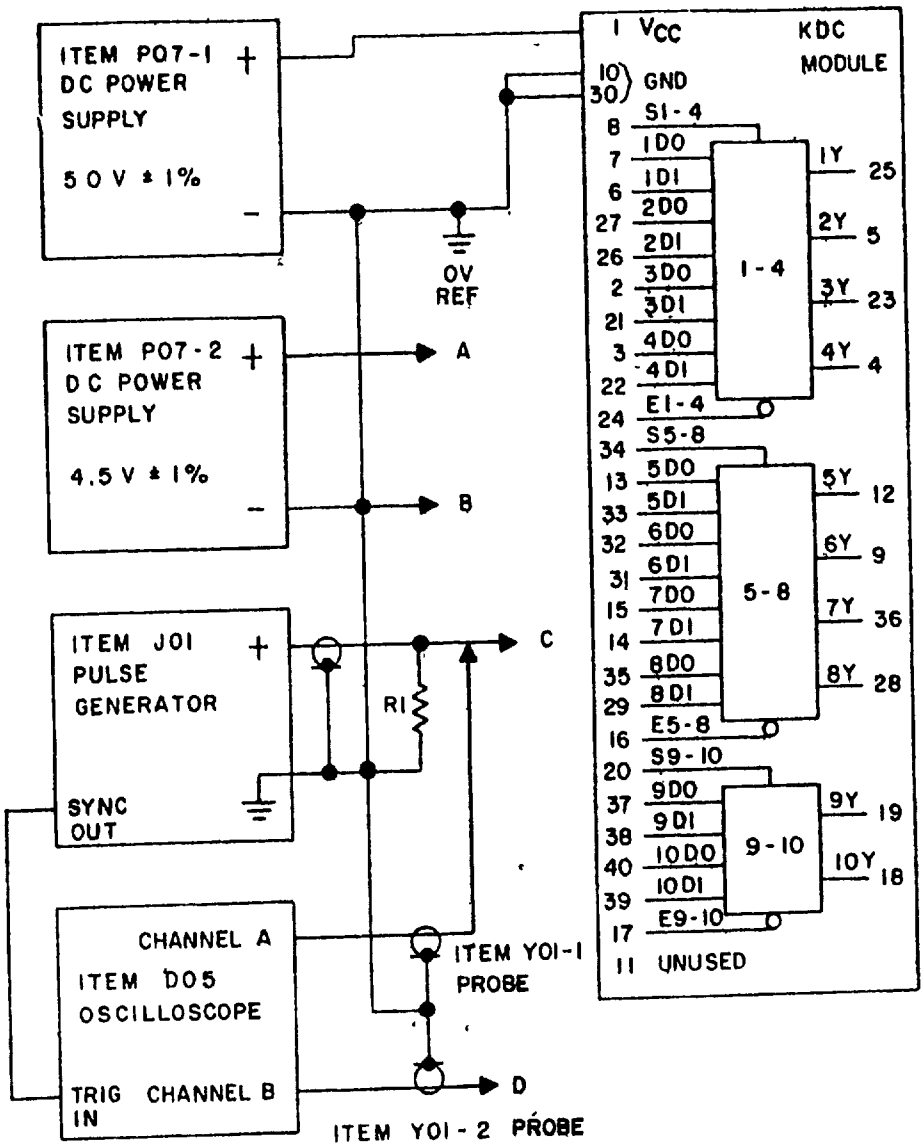
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U2	E1-4 to 2Y Propagation delay time (t_{PHL}), C_L = 50 pF	27	8	24	5	3	34	ns	16,19
U3	E1-4 to 3Y	2	8	24	23				
U4	E1-4 to 4Y	3	8	24	4				
U5	E5-8 to 5Y	13	34	16	12				
U6	E5-8 to 6Y	32	34	16	9				
U7	E5-8 to 7Y	15	34	16	36				
U8	E5-8 to 8Y	35	34	16	28				
U9	E9-10 to 9Y	37	20	17	19				
U10	E9-10 to 10Y	40	20	17	18				
U11	E1-4 to 1Y Propagation delay time (t_{PHL}), C_L = 150 pF	7	8	24	25		44		
U12	E1-4 to 2Y	27	8	24	5				
U13	E1-4 to 3Y	2	8	24	23				
U14	E1-4 to 4Y	3	8	24	4				
U15	E5-8 to 5Y	13	34	16	12				

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TABLE XIV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U16	E5-8 to 6Y Propagation delay time (t_{PHL}), C_L = 150 pF	32	34	16	9	3	44	ns	16,19
U17	E5-8 to 7Y	15	34	16	36				
U18	E5-8 to 8Y	35	34	16	28				
U19	E9-10 to 9Y	37	20	17	19				
U20	E9-10 to 10Y	40	20	17	18				

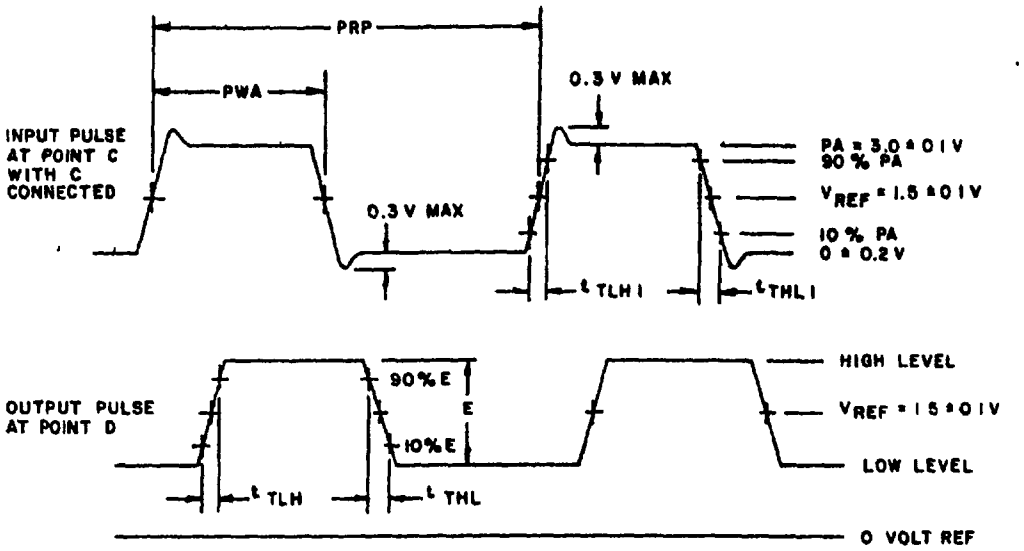
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NOTES:
1. R1 = 51.1Ω ±5%, 1 W.
2. Load each output under test with a load as specified on figure 12.

FIGURE 16. Transition time and propagation delay time test circuit for table XIV.

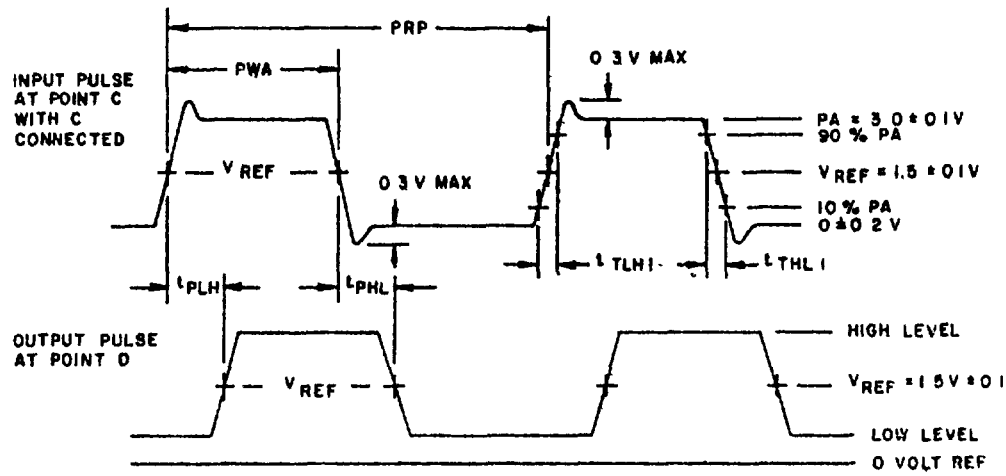
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:46Z
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. Before making the measurement specified in table XIV verify that the output pulse high levels is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 17. Transition time test circuit waveforms for table XIV.

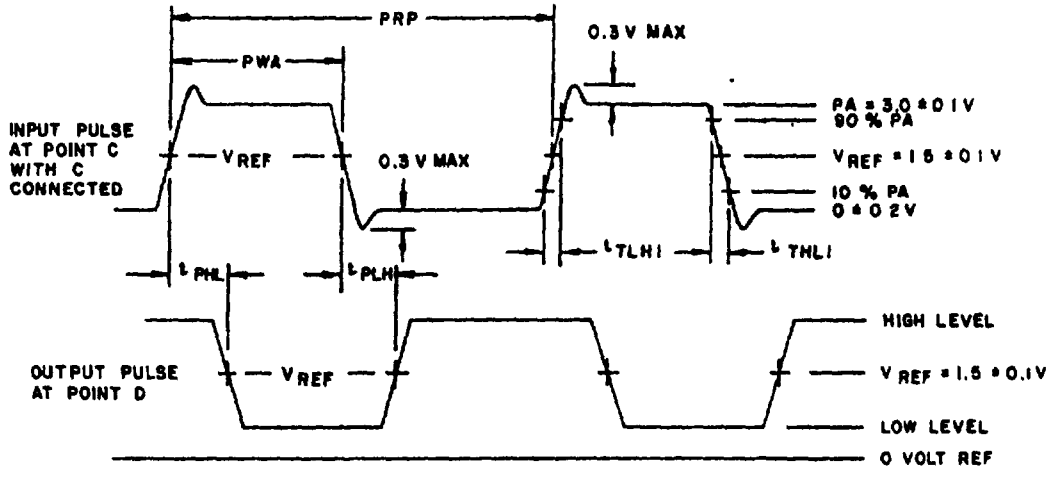
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XIV verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 18. Propagation delay time (data and select to Y output) test circuit waveforms for table XIV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XIV verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 19. Propagation delay time (enable to Y output) test circuit waveforms for table XIV.

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TABLE XV. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1Y Output crosstalk, low level	8	6	14,21,22,26,29, 31,33,38,39	25	0	1	V	20, 21
V2	2Y	8	26	6,14,21,22,29, 31,33,38,39	5				
V3	3Y	8	21	6,14,22,26,29, 31,33,38,39	23				
V4	4Y	8	22	6,14,21,26,29, 31,33,38,39	4				
V5	5Y	34	33	6,14,21,22,26, 29,31,38,39	12				
V6	6Y	34	31	6,14,21,22,26, 29,33,38,39	9				
V7	7Y	34	14	6,21,22,26,29, 31,33,38,39	36				
V8	8Y	34	29	6,14,21,22,26, 31,33,38,39	28				
V9	9Y	20	38	6,14,21,22,26, 29,31,33,39	19				
V10	10Y	20	39	6,14,21,22,26, 29,31,33,38	18				

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V11	1Y Output crosstalk, low level	8,34,20	6,26,21,22	33,31,14,29, 38,39	25	0	400	mV	20, 21
V12	2Y				5				
V13	3Y				23				
V14	4Y				4				
V15	5Y		33,31,14,29	6,26,21,22, 38,39	12				
V16	6Y				9				
V17	7Y				36				
V18	8Y				28				
V19	9Y		38,39	6,26,21,22, 33,31,14,29	19				
V20	10Y		38,30	6,26,21,22, 33,31,14,29	18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:46Z
Check the source to verify that this is the current version before use.

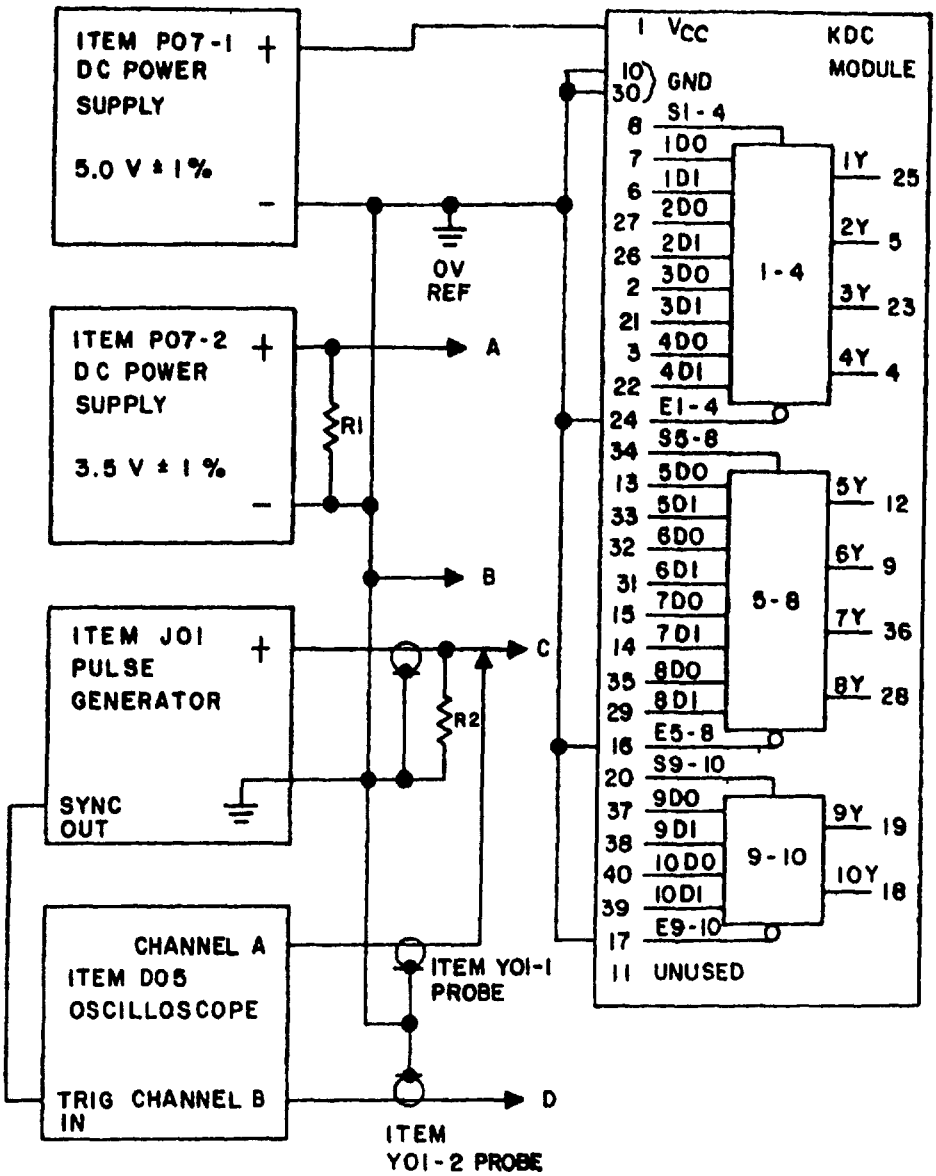
TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1Y Output crosstalk, high level	8,6	Open	14,21,22,26,29, 31,33,38,39	25	0	400	mV	20, 21
W2	2Y	8,26		6,14,21,22,29, 31,33,38,39	5				
W3	3Y	8,21		6,14,22,26,29, 31,33,38,39	23				
W4	4Y	8,22		6,14,21,26,29, 31,33,38,39	4				
W5	5Y	34,33		6,14,21,22,26, 29,31,38,39	12				
W6	6Y	34,31		6,14,21,22,26, 29,33,38,39	9				
W7	7Y	34,14		6,21,22,26,29, 31,33,38,39	36				
W8	8Y	34,29		6,14,21,22,26, 31,33,38,39	28				
W9	9Y	20,38		6,14,21,22,26, 29,31,33,39	19				
W10	10Y	20,39		6,14,21,22,26, 29,31,33,38	18				

TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	400	mV	22, 23

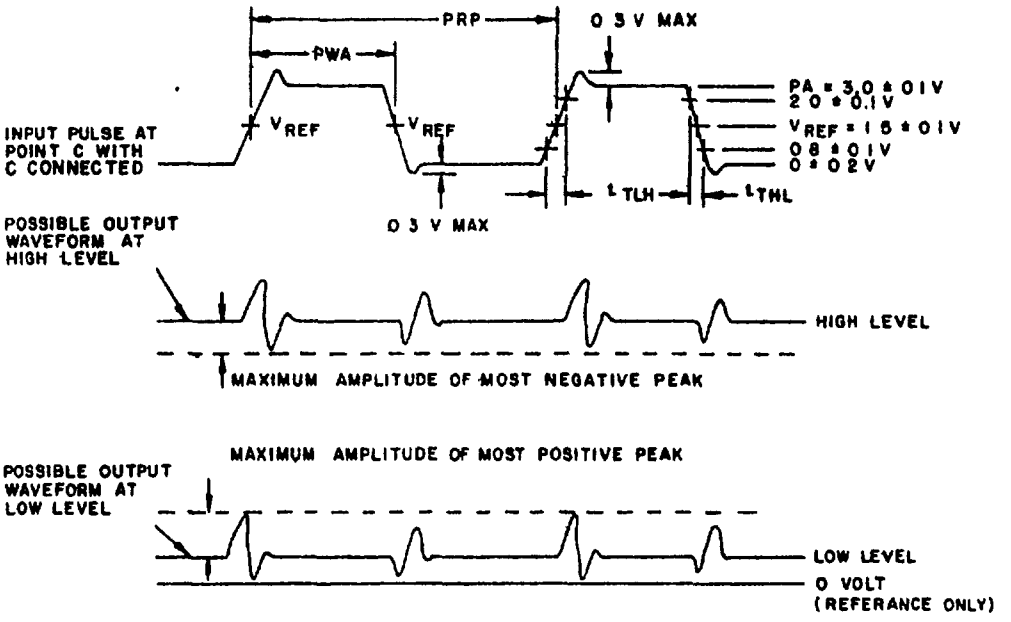
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 12.

FIGURE 20. Output crosstalk (low-level and high-level) test circuit for table XV.

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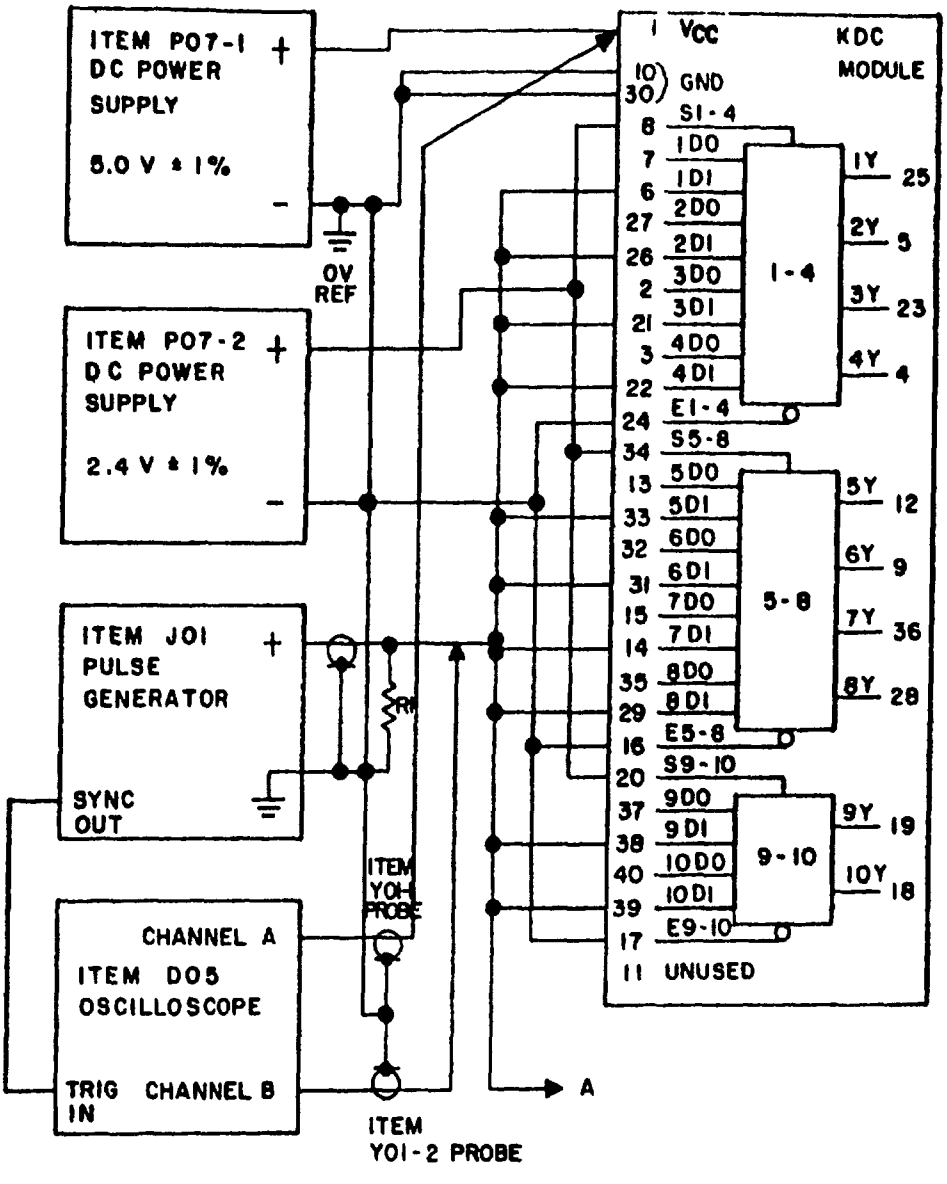


NOTES:

1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors shall be less than one inch (25.4 mm) between resistor R2 and input pins; probes Y01-1 and Y01-2 to input and output pins; load circuits to output pins; and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit waveforms for table XV.

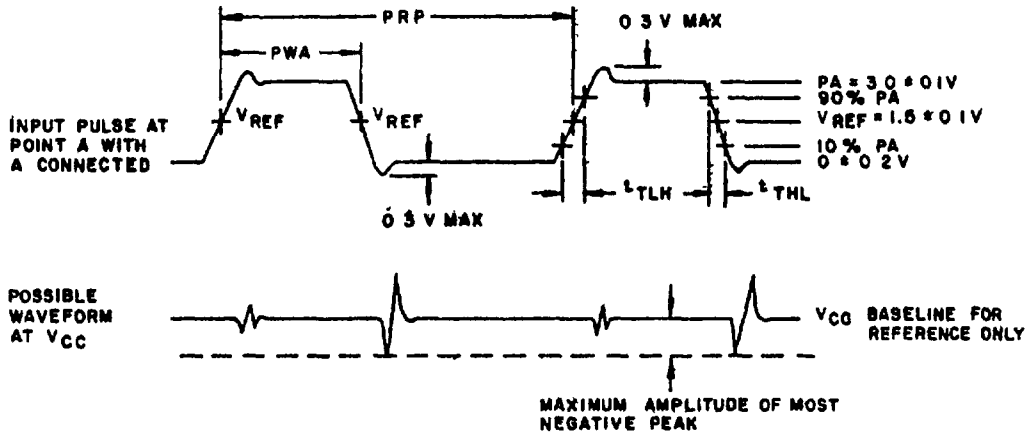
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NOTE: R1 = 51.1Ω ±5%, 1 W.

FIGURE 22. Power supply transient amplitude test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 1$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

FIGURE 23. Power supply transient amplitude test circuit waveforms for table XVI.

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TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply filter capacitance	Open	1	1.6	3.0	µF	24

TABLE XVIII. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Z1	1D0	Input capacitance	7	Open	2.0	15.0	pF	24
Z2	1D1		6					
Z3	2D0		27					
Z4	2D1		26					
Z5	3D0		2					
Z6	3D1		21					
Z7	4D0		3					
Z8	4D1		22					
Z9	5D0		13					
Z10	5D1		33					
Z11	6D0		32					
Z12	6D1		31					
Z13	7D0		15					
Z14	7D1		14					
Z15	8D0		35					
Z16	8D1		29					

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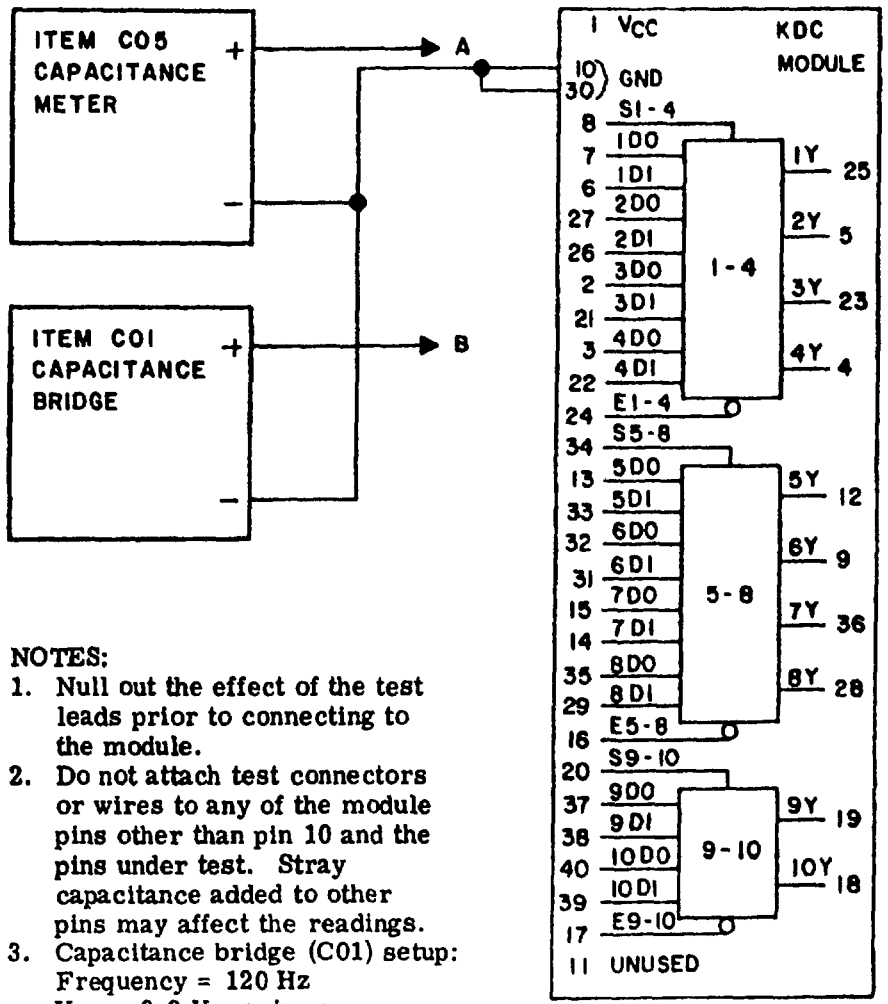
TABLE XVIII. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z17	9D0 Input capacitance	37	Open	2.0	15.0	pF	24
Z18	9D1	38					
Z19	10D0	40					
Z20	10D1	39					
Z21	S1-4	8					
Z22	S5-8	34					
Z23	S9-10	20					
Z24	E1-4	24					
Z25	E5-8	16					
Z26	E9-10	17					

TABLE XIX. Ground continuity.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	Ground continuity	10	30	0.0	1.0	Ω	25
AA2	Ground continuity	10	30	0.0	0.1	Ω	25

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connectors or wires to any of the module pins other than pin 10 and the pins under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup: Frequency = 120 Hz
   V_{pk} = 0.3 V maximum.

FIGURE 24. Power supply filter capacitance and input capacitance test circuit for tables XVII and XVIII.

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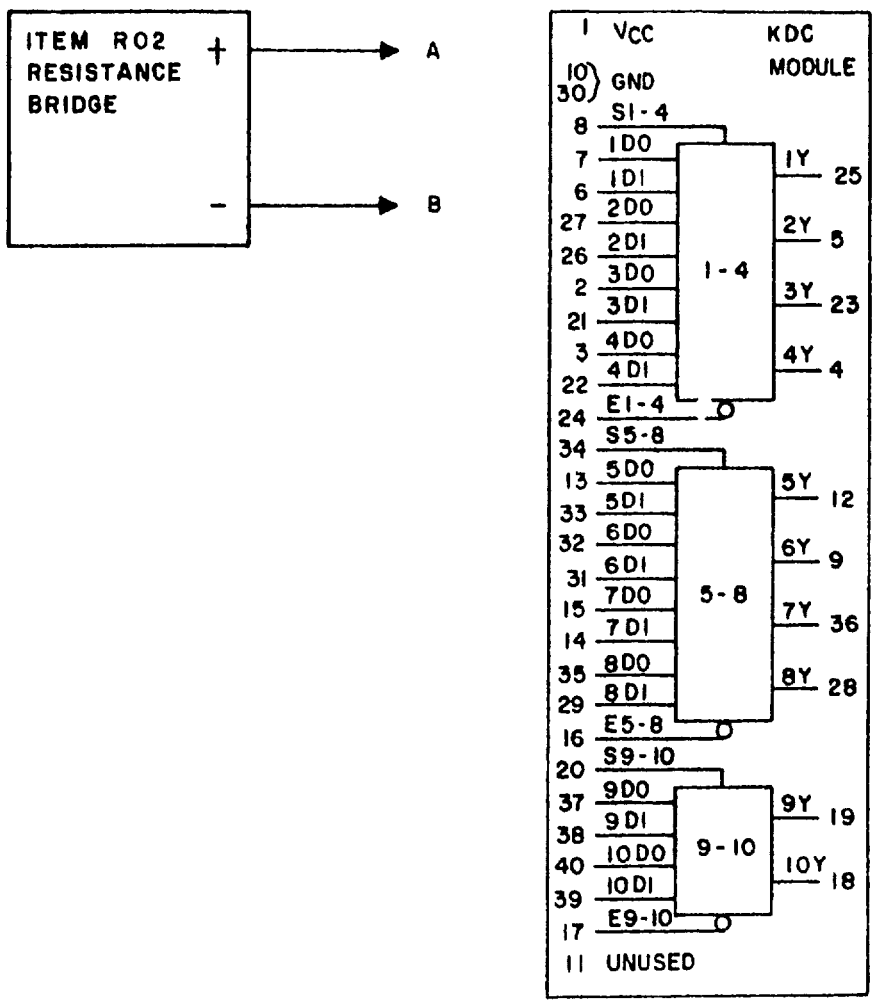


FIGURE 25. Ground continuity test circuit for table XIX.

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